Final Group Project

This final group project is the main integrative component of the course. The objective is not to write a theoretical paper, but to demonstrate that you can combine transversal knowledge developed at IE (in this course and others) into a coherent, feasible, and high-quality applied project. You will define your own topic, justify why it matters, select appropriate data and methods, and communicate results clearly to both technical and non-technical audiences.

Rubrics and detailed scoring guides for each component will be released later and shared on Blackboard. All following information is tentative.

Grades and timeline (tentative)

Component	Weight	Description	Due / when
First report (Project proposal)	10% (10/100)	Grant-style proposal to assess feasibility and direction	04/03/2026
Final report	20% (20/100)	Full project	15/04/2026
Group presentation	10% (10/100)	10-minute presentation + Q&A	Sessions 28/29 (23/04 and 27/04)

Total project weight: 40% of the final course grade.

General guidelines and expectations

- **Up to 6** team members per group. Groups are **self-selected**.
- All members must contribute meaningfully.
- An *author contributions statement* is mandatory in the reports (both first report and final report – who did what).
- Topic is **free** but must be feasible within the timeline and available resources.
- AI tools are allowed, but you remain responsible for correctness and interpretation; disclose meaningful AI use briefly in the final report (in appendix).

1) First report (Project proposal) → 10%

This is a project proposal, similar to what you would send to a grant committee. The goal is to clearly define what you plan to do and allow the professor to assess feasibility early and give feedback.

What I want to learn from your proposal: What problem you are studying, why it matters, what you plan to test, what you expect to find (hypotheses), what data you need and where you are expecting to find it, and how your team will execute the work.

Required structure:

- Title, team, and roles
  - Include a provisional author contributions table (may change in the final report). Must be a table with name + planned contribution, for all members (max half-page)
- Introduction
  - Problem statement and motivation
  - Research question(s) + scope
  - Who is the stakeholder or use case? What decision could your analysis support?
- Literature review
  - Cover both (i) the topic/domain and (ii) the methods you plan to use
- Methods you plan to use (can be one or more of the topics of the course or even outside the topics covered in class)

    - Methods description
    - Hypotheses / expected outcomes. What do you expect and why?
    - Data needs and acquisition plan
    - Feasibility assessment and potential limitations (e.g., data access, compute, …)
- References

Length and formatting:

- Maximum **4 pages** (excluding references)
- Any page after page 4 will be disregarded for grading
- Standard format:
    - Times New Roman, 11 pt
    - Normal margins/spacing (standard Word/Google Docs defaults; do not compress)
- Maximum 2 display items total in the main text (**figures + tables combined**)

Evaluation criteria (tentative – rubric will be shared in the coming sessions):

- Clarity and coherence (question → data → proposed methods)
- Quality/relevance of literature review
- Originality balanced with feasibility
- Realistic plan and clear roles
- Respect formatting requirements

2) Final report → 20%

This is the final, polished output of your project. I expect a professional, scientific-style report (take inspiration from Nature journals[1]) with clear methods, detailed results, and thoughtful discussion. The goal is not only to "train a model," but to answer a meaningful question and communicate credible evidence.

Required structure for the main text:

- Abstract
    - (**MAX 150 words**: problem → approach → key results → implication)
- Introduction
    - Research gap
    - The novelty of your approach/the area where you apply existing methods
- Results (show the most important findings first; use your best figures/tables)
- Discussion
    - Results interpretation
    - implications/recommendations
    - limitations
    - future work/directions
- Methods (what you did, clearly and reproducibly)
    - Which method(s) you used
    - Which data you used and where you collect them
- Author contribution
    - Final author contributions table (name + contribution, for all members - max half-page)

[1] https://www.nature.com/nature/for-authors/formatting-guide

Appendix structure:

- References
- Code availability statement (if you have a code, explain whether/how code can be run)
- AI usage statement

Length and formatting:

- Maximum **8 pages** for the main text (excluding appendix).
- Any page after page 8 will be disregarded for grading.
- Standard format:
  - Times New Roman, 11 pt
  - Normal margins/spacing (standard Word/Google Docs defaults; do not compress)
- Maximum 4 display items total in the main text (**figures + tables combined**)

Evaluation criteria (tentative – rubric will be shared in the coming sessions):

- Clarity and rigor of methods and evaluation
- Impact and credibility of results (not raw performance alone)
- Depth of discussion: uncertainty, limitations, generalization, ethics (when relevant)
- Quality and usefulness of figures/tables
- Concision and overall scientific communication quality
- Respect formatting requirements
- Reproducibility (explicitly assessed: 5 points). Your work must be reproducible either by:
  - sharing runnable code, or
  - providing a clear and complete step-by-step description of the pipeline (data → preprocessing → modeling → evaluation)
  - If code is shared, it should run (or clearly state what cannot run and why), and include basic instructions and environment notes in the appendix

3) Group presentation → 10%

Format

- 10 minutes strict timing
- 2 questions from the audience
- Visual aid allowed (slides/poster). Aim for clarity for a broad audience

What to cover (recommended)

- Problem and motivation (why it matters)
- Data (what you used; key constraints)
- Methods (what you tried; what you selected; why)
- Results (main findings, ideally 1–2 key visuals)
- Discussion (implications, limitations, next steps)

Grading

- 5 points: professor evaluation
- 5 points: peer evaluation (implementation details TBD)

Practical guidance

- Start early with a baseline and a clear evaluation plan.
- Choose a question where you can produce results that matter (a decision, an insight, a recommendation), not just a model.
- Keep the main report focused; use the appendix for supporting material.